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PG – 851

I Semester M.C.A. (2 Years Course) Degree Examination, August/September 2021
(CBCS Scheme) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA2 : Discrete Mathematics

Time : 3 Hours Max. Marks : 70

***Instruction** : Answer **any five** from Part – **A** and **any four** from Part – **B**.*

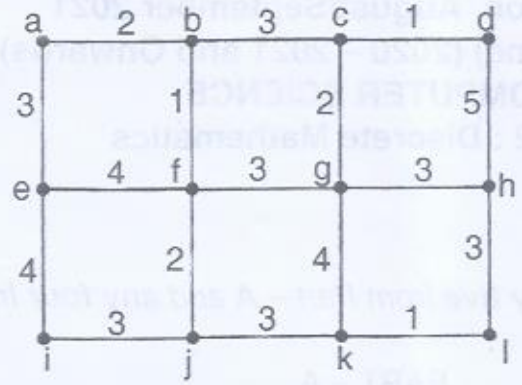
PART – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. a) If A, B and C are three sets, then prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. 3
   b) Prove that "null set is a subset of every set". 3
2. a) Define one-to-one and on-to functions with example. 3
   b) Show that $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$. 3
3. a) Let x be the set of factors of 12 and let $\leq$ be the relation divisor i.e., $x \leq y$, if and only if x divides y. Draw the Hasse diagram of $(x, \leq)$. 3
   b) Find the values of the extended binomial coefficients $\binom{-2}{3}$ and $\binom{1/2}{3}$. 3
4. Obtain an explicit form for the following sequence a_n defined recursively by $a_n = 2a_{n-1} + 1$ for $n \geq 2$, with $a_1 = 3$. 6
5. What is the expected value of the sum of the numbers that appear when a pair of fair dice is rolled ? 6
6. a) A sequence of 10 bits is randomly generated. What is the probability that at least one of these bits is 0 ? 3
   b) Let E_1 and E_2 be events in sample space S. Then show that $p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$. 3

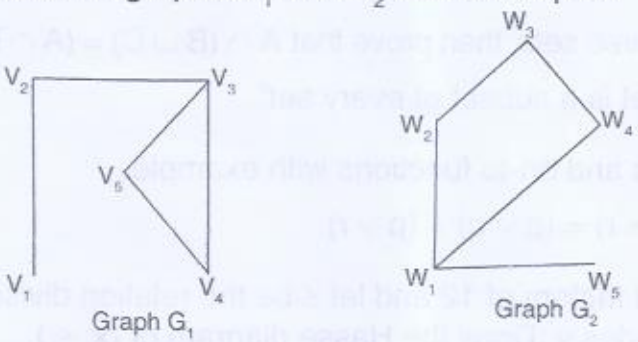
P.T.O.

7. Use Krushkal's algorithm to find a minimum spanning tree in the weighted graph given below. 6



8. a) Define a bipartite graph. Show that the hypercube Q_3 is a bipartite graph. 3

b) Check whether graphs G_1 and G_2 are isomorphic. 3



PART – B

Answer **any four** questions. **Each** question carries **10** marks. **(4×10=40)**

9. a) Prove that $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \Leftrightarrow \neg(q \vee p)$.

b) A survey among 100 students shows that of the three ice cream flavors Vanilla, Chocolate and Strawberry, 50 students like Vanilla, 43 like Chocolate, 28 like Strawberry, 13 like Vanilla and Chocolate, 11 like Chocolate and Strawberry, 12 like Strawberry and Vanilla and 5 like all of them. Find the number of students surveyed who like each of the following flavors.

i) Chocolate but not Strawberry.

ii) Chocolate and Strawberry but not Vanilla. **(5+5=10)**

10. a) By mathematical induction, prove that for every integer n, the number $A_n = 5^n + 2.3^{n-1} + 1$ is a multiple of B.

b) The matrix of a relation on the set A = {1, 2, 3} is given by $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

Show that R is an equivalence relation. **(5+5=10)**

11. a) Shirts numbered consecutively from 1 to 20 are worn by students of a class. When any 3 of these students are chosen to be debating team from the class, the sum of their shirt numbers is used as the code number of the team. Show that if any 8 of the 20 are selected, then from these 8 we may form at least two different teams having the same code number.

    b) A committee of eight people is formed from two mathematicians and ten economists. In how many ways can it be done, if the committee must include at least one mathematician ? **(5+5=10)**

12. a) State and prove the Bayes' theorem.

    b) The probability distribution of a discrete random variable is given below :

x	–2	–1	0	1	2	3
P(x)	0.1	K	0.2	2K	0.3	K

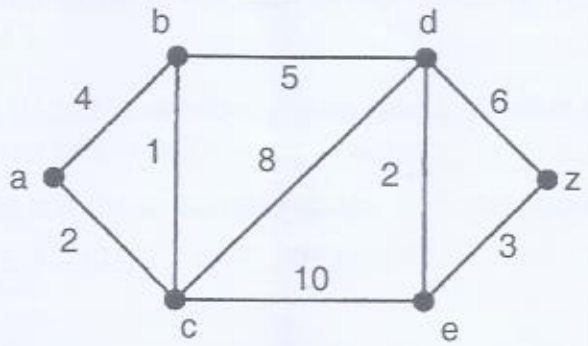
    Find : i) K ii) Mean and iii) Variance. **(5+5=10)**

13. a) Two cards are drawn from a pack of cards at random. What is the probability that it will be
        i) a diamond and a heart
        ii) a king and a queen
        iii) two kings ?

    b) State and prove the first theorem in Graph theory. Further, prove that the number of vertices of odd degree in a graph is always even. **(5+5=10)**

14. a) Define an Eulerian graph. If G is an Eulerian graph, then show that all the vertices of G are of even degree.

    b) Use Dijktra's algorithm, find the shortest path between a to z in the weighted graph.



(5+5=10)

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I Semester M.C.A. (2 Years Course) Examination, July 2022
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA2 : Discrete Mathematics

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **any 5** questions from Part – **A**.*
*2) Answer **any 4** questions from Part – **B**.*

PART – A

Answer **any five** questions. **Each** question carries **six** marks.

1. a) If A, B, C are three sets, then prove that $(A \cup B) \cup C = A \cup (B \cup C)$. **3**
   b) Determine the sets A and B, given that $A - B = \{1, 3, 7, 11\}$, $B - A = \{2, 6, 8\}$, $A \cap B = \{4, 9\}$. **3**

2. Let $A = \{1, 2, 3, 4, 6\}$ and R be the relation on A defined by aRb if and only if "a is a multiple of b".
   i) Write down R as a set of ordered pair.
   ii) Represent R as a matrix.
   iii) Draw the diagraph of R. **6**

3. a) Let $f : R \to R$ is defined by $f(x) = 2x + 5$ prove that f is one-one and onto. **3**
   b) Prove that the statement $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$. **3**

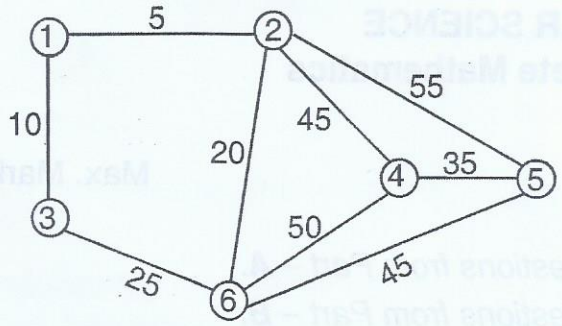
4. A sequence $\{a_n\}$ is defined recursively by $a_1 = 4$, $a_n = a_{n-1} + n$ for $n \geq 2$. **6**

5. a) In how many ways can the letters of the word "MATHMATICS" be arranged such that the words starts and ends with M ? **3**
   b) Find the number of triangles formed by joining the vertices of a polygon of 12 sides. **3**

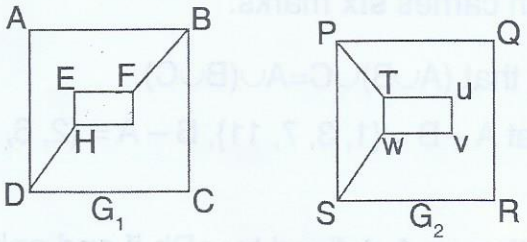
6. What is the expected value of the sum of the numbers that appears when a pair of fair dice is rolled ? **6**

P.T.O.

7. Find minimum weight spanning tree by Kruskal's algorithm. 6



8. Examine whether the following pair of graphs are isomorphic or not. Justify your answer. 6



PART – B

Answer **any four** questions. **Each** question carries **10** marks.

9. a) In a survey of 260 college students, 64 had taken a Mathematics course, 94 had taken a Computer Science, 58 had taken a Business course, 28 had taken both Mathematics and Business courses, 26 had taken both Mathematics and Computer Science courses, 22 had taken both Computer Science and Business courses and 14 had taken 3 types of courses.
   i) How many of these students had taken none of the 3 courses ?
   ii) How many had taken only a Computer Science course ?

   b) Let A, B and C be three finite sets prove that
   $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$. **(5+5)**

10. a) By mathematical induction, prove that the sum of cubes of 'n' natural numbers is equal to $\frac{n^2(n+1)^2}{4}$ for all 'n' natural numbers.

b) The matrix of a relation on the set A = {1, 2, 3} is given by $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Show that R is an equivalence relation. **(5+5)**

11. a) Prove that for any propositions p, q and r the compound proposition $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology.

b) If a band could not play rock music or the refreshments were not delivered on time, then the new year party would have been cancelled and Alicia would have been angry. If the party were cancelled, the refunds would have to be made. No refunds were made. Therefore, the band could play rock music. Establish the validity of the argument by using the rules of inferences. **(5+5)**

12. a) State and prove the extended pigeon hole principle.

b) The probability distribution of a discrete random variable is given below.

X	–2	–1	0	1	2	3
P(X)	0.1	K	0.2	2K	0.3	K

Find (i) K (ii) Mean (iii) Variance. **(5+5)**

13. a) Three coins are tossed in succession. Find out the probabilities of occurrences of

i) Two consecutive heads

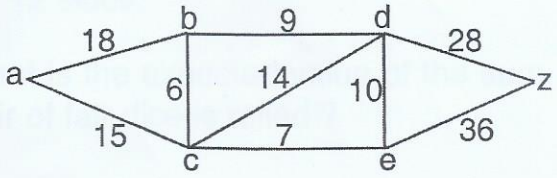
ii) Two heads and

iii) Two heads in the following order, head, tail and head.

b) State and prove Hand Shaking theorem. **(5+5)**

14. a) Prove that, in any undirected graph, the number of odd degree vertices is even.

b) Using Dijktra's algorithm, find the shortest path between a to z in the weighted graph. **(5+5)**



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I Semester M.C.A. (Two Years Course) Examination, July 2023
(CBCS) (2020 – 2021 and Onwards)
COMPUTER SCIENCE
1MCA 2 : Discrete Mathematics

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** from Part – **A** and **any four** from Part – **B**.*

PART – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. a) Prove that, for any three sets A, B and C
$A \times (B \cup C) = (A \times B) \cup (A \times C)$. **3**

   b) Prove that "Null set is a subset of every set". **3**

2. a) Find inverse, converse and contrapositive of the following implication. "If the weather is sunny, then I will go to school." **3**

   b) Let x be the set of factors of 12 and let "≤" be the relation divisor i.e., $x \leq y$ if and only if x divides y. Draw the Hasse diagram of $(X, \leq)$. **3**

3. a) Let $f : R \to R$, determine whether f is invertible and if so, determine f^{-1} where $f = \{(x, y) \mid 2x = 3y = 7\}$. **3**

   b) Show that $p \to (q \vee r) \equiv (p \to q) \vee (p \to r)$. **3**

4. Obtain an explicit form for the following sequences $\{a_n\}$ defined recursively by $a_n = 2a_{n-1} + 1$ for $n \geq 2$ with $a_1 = 3$.

5. What is the expected value of the sum of the numbers that appear when a pair of fair dice is rolled ?

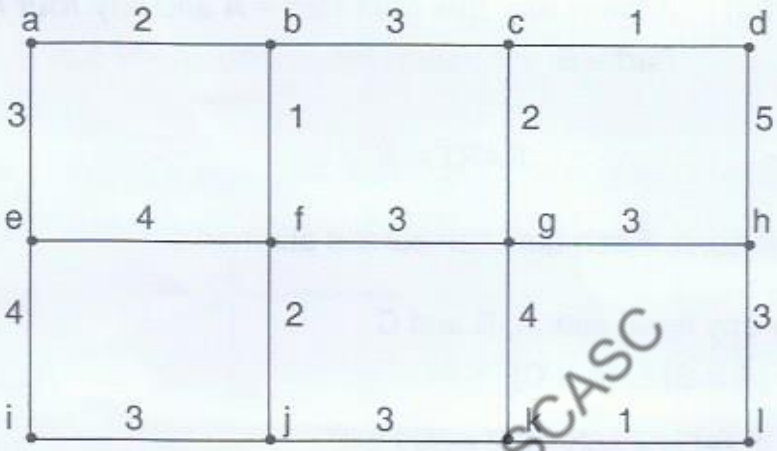
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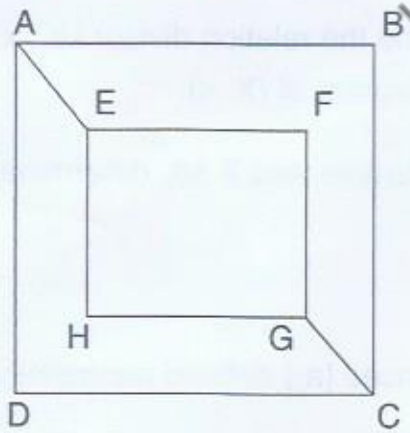
6. a) How many 5-digit telephone numbers can be constructed using the digits 0 to 9, if each number starts with 67 and no digit appears more than once ? **3**

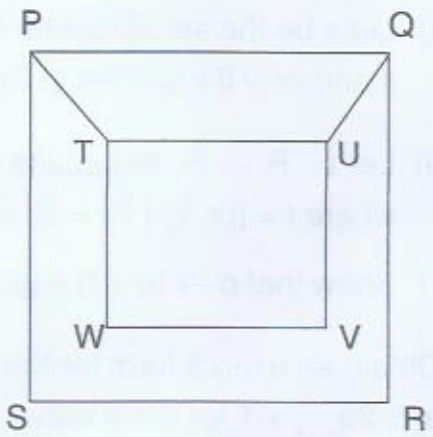
   b) In how many ways a five member committee can be formed out of ten people if two particular people must be included ? **3**

7. Use Kruskal's algorithm to find a minimum spanning tree in the weighted graph given below.



8. Examine whether the following pair of graphs are isomorphic or not. Justify your answer.





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PART – B

Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9. a) There is a total of 200 students in Class XI. 120 of them study Mathematics, 50 students study Commerce and 30 students study both Mathematics and Commerce. Find the number of students who
   i) Study Mathematics but not Commerce.
   ii) Study Commerce but not Mathematics.
   iii) Study Mathematics or Commerce. **5**

   b) Prove that $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \Leftrightarrow \neg(q \vee p)$. **5**

10. a) By mathematical induction, prove that the sum of cubes of 'n' natural numbers is equal to $\frac{n^2(n+1)^2}{4}$ for all 'n' natural numbers. **5**

    b) Define a relation R on set A = {1, 2, 3, 4, 5} by $(x, y) \in R$ if R = {(a, b) : |a – b| is even}, then show that R is an equivalence relation. **5**

11. a) Prove that for any proposition p, q, r, the following compound proposition $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology. **5**

    b) A committee of 8 people is formed from two mathematicians and ten economists. In how many ways can it be done, if the committee must include at least one mathematician ? **5**

12. a) State and prove Baye's theorem. **5**

    b) The probability distribution of a discrete random variable is given below : **5**

X	– 2	– 1	0	1	2	3
P(X)	0.1	K	0.2	2K	0.3	K

Find :
i) K
ii) Mean
iii) Variance.

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13. a) Three coins are tossed in succession. Find out the probabilities of occurrences.

i) two consecutive heads

ii) two heads and

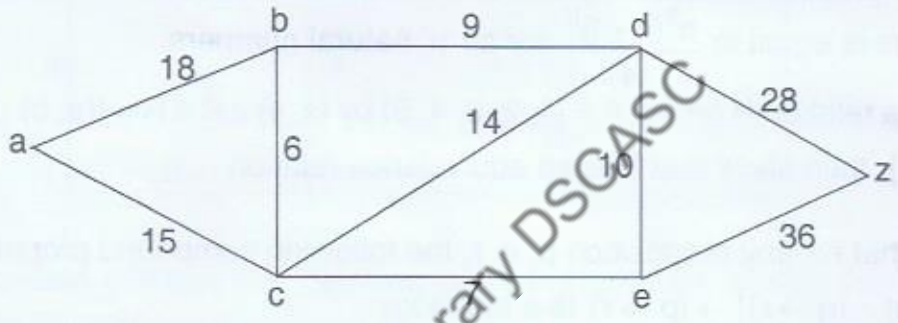
iii) two heads in the following order :

head, tail, head. **5**

b) State and prove Hand Shaking theorem. **5**

14. a) Prove that in any undirected graph, the number of odd degree vertices is even. **5**

b) Using Dijkstra's algorithm, find the shortest path between a to z in the weighted graph. **5**



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I Semester M.C.A. (Two Years Course) Examination, June/July 2024
(CBCS) (2020-2021 and Onwards)
COMPUTER SCIENCE
1MCA2 : Discrete Mathematics

Time : 3 Hours Max. Marks : 70

Instruction** : Answer **any 5** from Section – **A** and **any 4** from Section – **B.

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks : **(5×6=30)**

1. Apply principle of mathematical induction to test, for every positive integer n, 5 divides $n^5 - n$.

2. Show that, the compound proposition, $[(p \wedge \neg q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)]$ is a tautology.

3. Calculate the number of arrangements for all letters in the word SOCIOLOGICAL. In how many of these arrangements
   a) A and G are adjacent ?
   b) All the vowels are adjacent ?

4. For the Fibonacci sequence $F_0, F_1, F_2, \ldots$, examine $F_n =$

5. The probability of a person that will get an electrification contract is 2/5 and the probability of not getting a plumbing contract is 4/7. If the probability that he will get not less than 1 contract is 2/3, find the probability of the event that he will get both contracts.

6. What is the probability of obtaining 45 or fewer heads in 100 tosses of a coin ?

7. Define the following with example :
   Graph, Adjacency matrix, Connected graph, Weighted graph.

8. Show that a tree with n vertices will consist of n–1 edges.

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P.T.O.

SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks : **(4×10=40)**

9. Answer the following :

   a) In a college, 20 professors teach mathematics or physics. If 12 teach mathematics and 4 teach both physics and mathematics, how many teach physics ?

   b) Show that $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$ using the laws of logic.

10. Answer the following :

   a) A student has to answer 10 questions, choosing at least 4 from each of Parts A and B. If there are 6 questions in Part A and 7 in Part B, in how many ways can the student choose 10 questions ? How many numbers are there between 99 and 1000 having at least one of their digits 7 ?

   b) The number of virus affected files in a system is 1000 (to start with) and this increases 250% every two hours. Use a recurrence relation to determine the number of virus affected files in the system after one day.

11. A card has been drawn from a well-shuffled deck of 52 cards. What will be the probability that a card will be an

   a) Diamond

   b) Black card

   c) Not an ace

   d) Not a diamond.

12. A tyre manufacturing company kept a record of the distance covered before a tyre needed to be replaced. The table shows the results of 1000 cases. If a tyre is bought from this company, what is the probability that :

   a) It has to be replaced before 4000 km is covered ?

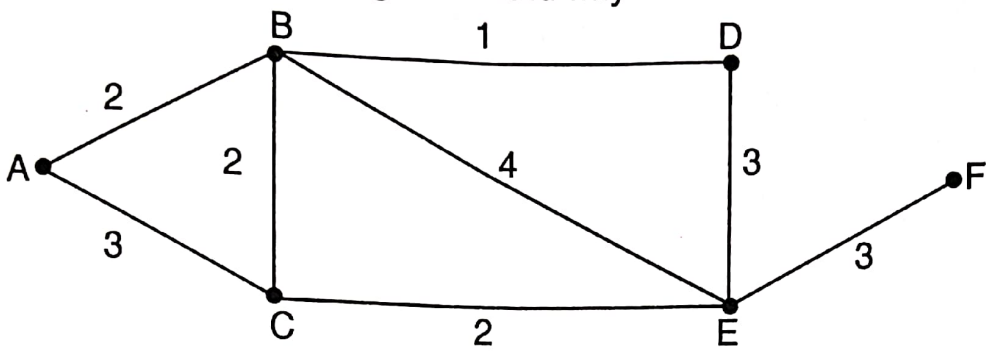
   b) It will last more than 9000 km ?

   c) It has to be replaced after 4000 km and 14000 km is covered by it ?

Distance (in km)	< 4000	4000 to 9000	9001 to 14000	> 14000
Frequency	20	210	325	445

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13. Find MSP for the following graph using Prim's and Kruskal's algorithm. Which is the efficient algorithm and why ?



14. Answer the following :

a) Let A = {1, 2, 3, 4, 6, 12}. On A, define the relation R by aRb if and only if a divides b. Prove that R is a partial order on A. Draw the Hasse diagram for this relation.

b) ABC is an equilateral triangle whose sides are of length 1 cm each. If we select 5 points inside the triangle, show that the distance between them is less than ½ cm.

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I Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
1MCA2 : Discrete Mathematics
Equivalent to Discrete Mathematics and its Application (2013 and Below)

Time : 3 Hours | Max. Marks : 70

Instructions : *1) Answer **any 5** from Section – **A**.*
*2) Answer **any 4** from Section – **B**.*

SECTION – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. a) Prove that "Null set is a subset of every set". **3**

   b) If A, B and C are three sets, then prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. **3**

2. Show that the compound propositions $[p \rightarrow (\sim q \vee r)] \leftrightarrow \sim[p \rightarrow (q \rightarrow r)]$ is contradiction or not. **6**

3. a) Let $f : R \rightarrow R$ is defined by $f(x) = 7x + 5$. Prove that f is one-one and onto function. **3**

   b) Let A = {1, 2, 3, 4, 6} and R be the relation on A defined by aRb if and only if "a is a multiple of b". **3**
      i) Write down R as a set of ordered pair.
      ii) Represent R as a matrix form.
      iii) Draw the digraph of R.

4. Obtain an explicit form for the following sequence a_n defined recursively by $a_n = 2a_{n-1} + 1$ for $n \geq 2$ with $a_1 = 3$. **6**

5. a) Show that the open interval (0, 1) is uncountable. **3**

   b) Find converse, inverse and contrapositive of the following implication : "If it rains then the match will be cancelled." **3**

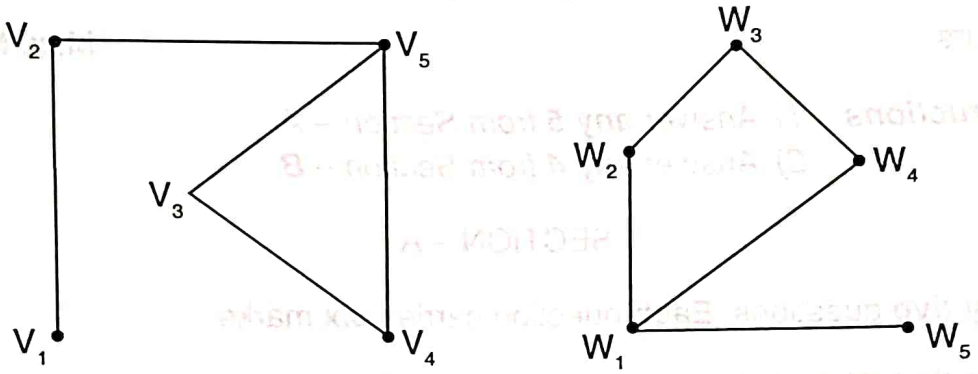
6. Define the following with example.
   Graph, Connected graph, Undirected Graph. **6**

P.T.O.

7. What is the expected value of the sum of the numbers that appear when a pair of dice is rolled ? 6

8. a) Define Bipartite graph with example. 3

b) Check whether the graphs G_1 and G_2 are Isomorphism. 3



SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

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9. a) In a survey of 200 students, it was found that 120 students study Mathematics, 90 students study Physics and 70 students study Chemistry, 40 Students study Mathematics and Physics, 30 students study Physics and Chemistry, 50 students study Chemistry and Mathematics and 20 students study none of the subjects. Find the number of students who study all the three subjects. 5

b) Define :

1) Modus Tollens

2) Syllogism

3) Modus Ponem. 5

10. a) By Mathematical Induction, prove that $1^3 + 2^3 + 3^3 + ... n^3 = \frac{n^2(n+1)^2}{4}$; where n is a positive integer. 5

b) The matrix of a relation on the set A = {1, 2, 3} is given by $M_n = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

Show that R is an equivalence relation. 5

11. a) State and prove Baye's Theorem. 5

b) The probability distribution of a Discrete Random Variable is given below. 5

x	–2	–1	0	1	2	3
P(x)	0.1	K	0.2	2K	0.3	K

12. a) Prove that for any propositions p, q, r, the compound proposition $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology. 5

b) Explain the Basic Logical Connectives. 5

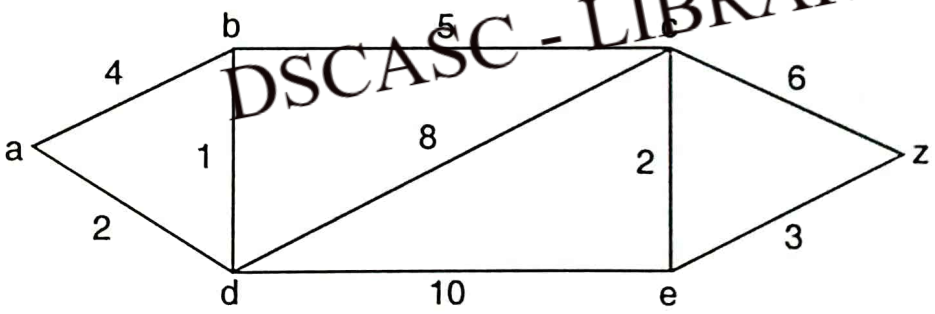
13. a) State and prove Hand Shaking Lemma. 5

b) Three coins are tossed in succession. Find out the probabilities of occurances 5

1) Two consecutive heads

2) Two tails.

14. By using Kruskal's and Prim's Algorithm, find the minimum spanning tree. 10



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